

Juan Carlos Ramirez Espinosa

SUMMARY

Mechatronics Engineer with experience in automation, cloud-related technologies, and data-driven solutions. Skilled in **AWS, Terraform, Python, and Linux**, with hands-on experience supporting enterprise environments, developing automated processes, and implementing practical solutions that improve efficiency and reliability.

EXPERIENCE

TATA Consultancy Services (TCS) - Site Reliability Engineer

01/2025 - Current

- Troubleshoot deployment issues using Terraform Cloud, Linux, and AWS services to enhance reliability and maintain operational continuity.
- Developed and supported automated installation workflows for ESP agents on Linux EC2 instances using AWS Systems Manager documents and Terraform.
- Streamlined automation for Stonebranch and ESP agent deployment and configuration, enhancing consistency and minimising manual setup tasks.
- Built dynamic SSM Command documents from templates with environment-specific variables to standardize agent deployments across multiple AWS environments.
- Integrated Ansible into SSM deployment workflows, executing post-installation configuration tasks on target instances to ensure compliance with operational standards.
- Supported secure payload delivery by implementing solutions based on IAM roles and restricted S3 bucket access, removing dependency on public object exposure.

Filtración del Bajío S.A. de C.V. - Maintenance Assistant

05/2015 - 07/2023

- Automated document ingestion via email using Amazon SES and S3; included sender validation, attachment extraction, PDF/XML matching, classification, and duplicate detection processes.
- Implemented AWS staging architecture with EC2, RDS PostgreSQL, S3, SES, Systems Manager, IAM, and Linux; secured containerized FastAPI REST API supported inventory management, movement tracking, invoice processing, voucher handling, catalog management, dashboard generation, and health checks.
- Developed Python-based platform for processing PDF/XML invoices and warehouse documents, applying inventory business rules, generating auditable movements, and automating operational reports.

PROJECTS

MIGSO - Project Engineer

08/2024 - 10/2024

- Built an employee portal using **Power Apps** and **SharePoint**, integrating user interfaces and earning selection for company implementation.

CERTIFICATIONS

Certification: Python

- HackerRank Certification | 2025

Certification: C#

- HackerRank Certification | 2025

Diploma: Data Analytics and Artificial Intelligence Tools.

- Instituto Tecnológico y de Estudios Superiores de Monterrey | 2023

Diploma: Professional Path of e-Series Universal Robots.

- Instituto Tecnológico y de Estudios Superiores de Monterrey | 2022

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EDUCATION

12/2024

Instituto Tecnológico y de Estudios Superiores de Monterrey

Querétaro, México

Bachelor of Science: Mechatronics

LINKEDIN

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SKILLS

- Cloud Computing: AWS Cloud Practitioner, Terraform, Harness
- Infrastructure as Code
- Linux administration
- Monitoring solutions
- Infrastructure monitoring
- Automation frameworks
- Automation & Control: TIA Portal, PLC fundamentals
- Programming: Python, C#, SQL, MATLAB, Unqork
- Design development